

Initially  $C_{\tau_i} = 1.42$ ,  $\gamma_i = 1.22$

Assume a value for  $\frac{A_e}{A^*} \rightarrow$  online read  $M_e > 1$   
Calc

online read  $\frac{P_e}{P_0}$  put in  $C_{\tau}$  Eq<sup>n</sup>, if matches with  $C_{\tau}$   
Calc

Value, correct guess, otherwise iterate!

After iterations,  $\frac{A_e}{A^*} = 1.67 \rightarrow M_e = 1.894$

h) If the initial chamber pressure is 1.5 MPa, what is the throat area ( $A^*$ ) of the nozzle?

Solution:

$$C_{\tau_i} = \frac{\tau_i}{P_0 A^*} \Rightarrow A^* = \frac{\tau_i}{P_0 C_{\tau_i}} = \frac{25560}{1.5 \times 10^6 \times 1.42} = 0.012 \text{ m}^2 = \boxed{120 \text{ cm}^2}$$

$$R^* = R_t = \sqrt{\frac{A^*}{\pi}} = \boxed{6.18 \text{ cm}}$$

i) what is the HRE burn time?

Solution:

$$R = \left\{ a(2n+1) \left( \frac{\dot{m}_{ex}}{\pi} \right)^n t + R_i^{2n+1} \right\}^{\frac{1}{2n+1}}$$

$$9 = \left\{ 1.6086(2.362) \left( \frac{6.283}{\pi} \right)^{0.681} t_b + 5^{2.362} \right\}^{\frac{1}{2.362}}$$

$$\Rightarrow t_b = 22.1 \text{ (s)}$$

(P-5)

Comment:

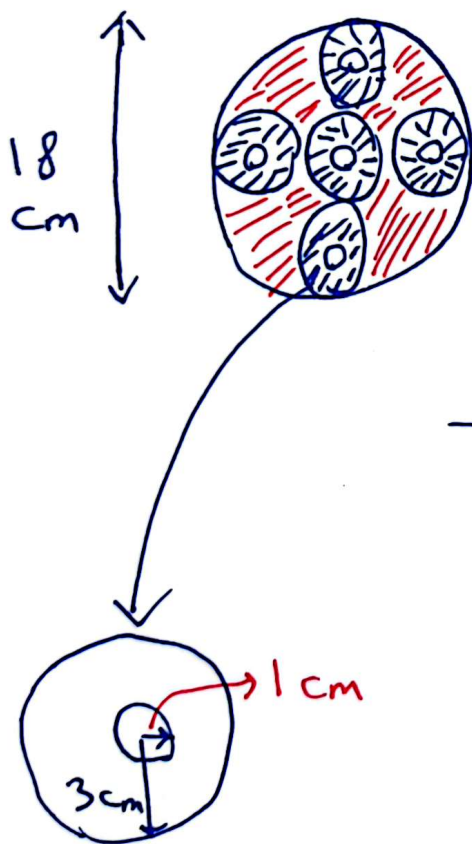
The length of the rocket is  $5\text{m} = 16.4\text{ft}$ , but the thrust is small  $25.56\text{ kN} = 5746\text{ lb}_f$

$$\text{If } M_o = \frac{T}{g} = \frac{25.56 \times 10^3}{9.81} = 2.61 \times 10^3\text{ kg}$$

$$\text{If } T = 2.9 M_o g \Rightarrow M_o = \frac{T}{2.9 g} = \frac{2.61 \times 10^3}{2.9} = \underline{900\text{ kg}}$$

To reduce the length and increase thrust per length, multiple ports are used.

Reconsider the previous example with the same outside radius of 9 cm, but with 5 ports of equal ~~size~~ cross-section.



// : Fuel (HTPB)

// : sliver

$$D_{\text{Rocket}} = 18\text{ cm} \rightarrow D_{\text{outer port}} = \frac{18}{3} = 6\text{ cm}$$

$$\rightarrow R_{\text{outer port}} = 3\text{ cm} \rightarrow R_{\text{port}} = 1\text{ cm}$$

Fuel thickness = 2 cm per port

(P-1)

oxidizer flow is equally split between ports

$$\dot{m}_{ox} = \frac{6.283}{5} = 1.257 \frac{\text{kg}}{\text{s}}$$

$$G_{ox,i} = \frac{\dot{m}_{ox}}{\pi R_i^2} = \frac{1.257}{\pi \times 1^2} = 0.4 \frac{\text{kg}}{\text{cm}^2 \text{s}}$$

$$r_i = 1.6086 (0.4)^{0.681} = 0.862 \frac{\text{cm}}{\text{s}} \quad \text{multi-port} > 0.288 \frac{\text{cm}}{\text{s}} \quad \text{single port}$$

$$\dot{m}_{f,i} = g_f 2\pi R_i L r_i = 0.915 \frac{\text{g}}{\text{cm}^3} \times 2\pi \times 1 \text{ cm} \times L \times 0.862 \frac{\text{cm}}{\text{s}}$$

$$\rightarrow \dot{m}_{f,i} = 4.956 L \frac{\text{g}}{\text{s cm}}$$

$$\uparrow \text{Single port } \dot{m}_{f,i} = 4139 \frac{\text{g}}{\text{s}}$$

$$\text{keep } \dot{m}_{f,i} \text{ the same} \Rightarrow 4139 \frac{\text{g}}{\text{s}} = 4.956 L \frac{\text{g}}{\text{s cm}}$$

$$\Rightarrow L = 167 \text{ cm} \quad \frac{2}{3} \text{ reduction in length from 500 cm!}$$

Let's compute final values of the rocket

$$G_{ox,f} = \frac{\dot{m}_{ox}}{\pi R_f^2} = \frac{1.257}{\pi \times 3^2} = 0.0445 \frac{\text{kg}}{\text{cm}^2 \text{s}}$$

final

$$r_f = 1.6086 \underbrace{(0.0445)^{0.681}}_{G_{ox,f}} = 0.193 \frac{\text{cm}}{\text{s}}$$

(P-2)

$$\dot{m}_{f,f} = \dot{m}_f (2\pi R_f L) v_f = 0.915 \times 2 \times \pi \times 3 \times (167 \times 0.193) = 556 \frac{\text{g}}{\text{s}} = 0.556 \frac{\text{kg}}{\text{s}}$$

$\dot{m}_{f,f}$   $\swarrow$   $\searrow$   
 $\dot{m}_{\text{fuel}}$   $\dot{m}_{\text{find}}$

$$\frac{\dot{m}_{\text{ox}}}{\dot{m}_{f,f}} = \frac{1.257}{0.556} = 2.26$$

From the table:  $c_f^* = 586$ ,  $\frac{f_t}{s} = 1789 \frac{\text{m}}{\text{s}}$   
 $\gamma_f = 1.14$   $\bar{M}_f = 22.3$

Now, we compute find time data using the same formula as used for initial time calculations:

$$T_{o,f} = 3479 \text{ K (from } c^* \text{ Eq.)}$$

$$\frac{A_e}{A_t}_f = 1.67 \quad \gamma = 1.14 \longrightarrow M_{e,f} = 1.85 \quad c_{\tau,f} = 1.42$$

$$u_{eq,f} = c_f^* c_{\tau,f} = 2540 \frac{\text{m}}{\text{s}}$$

$$I_{sp,f} = \frac{u_{eq,f}}{g_e} = 259 \text{ s}$$

$$R = \left\{ a(2n+1) \left( \frac{\dot{m}_{\text{ox}}}{\pi} \right)^n t_b + R_i \right\}^{\frac{1}{2n+1}}$$

$$R_f = 3 = \left\{ 1.6086 (2.362) \left( \frac{1.257}{\pi} \right)^{0.681} t_b + 1 \right\}^{\frac{1}{2.362}}$$

Solve  $t_b = 6.1 \text{ s} < 22.1 \text{ s}$   
 multi-part single part